

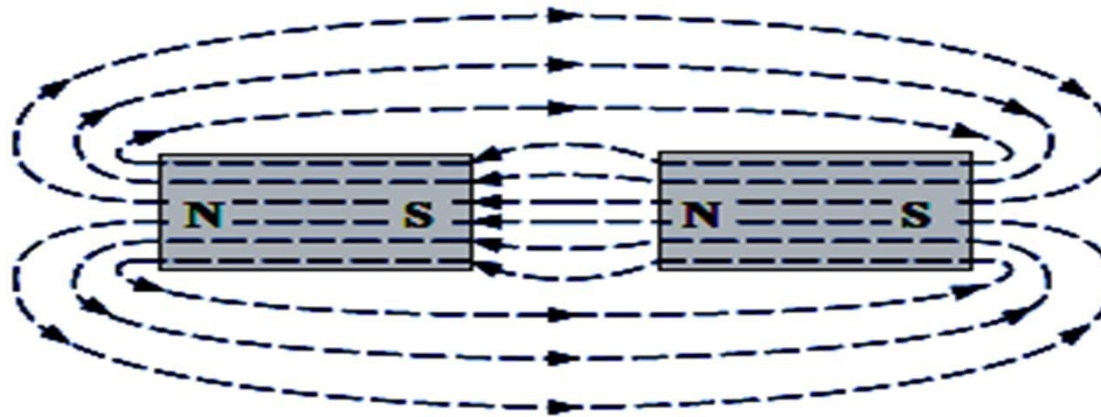
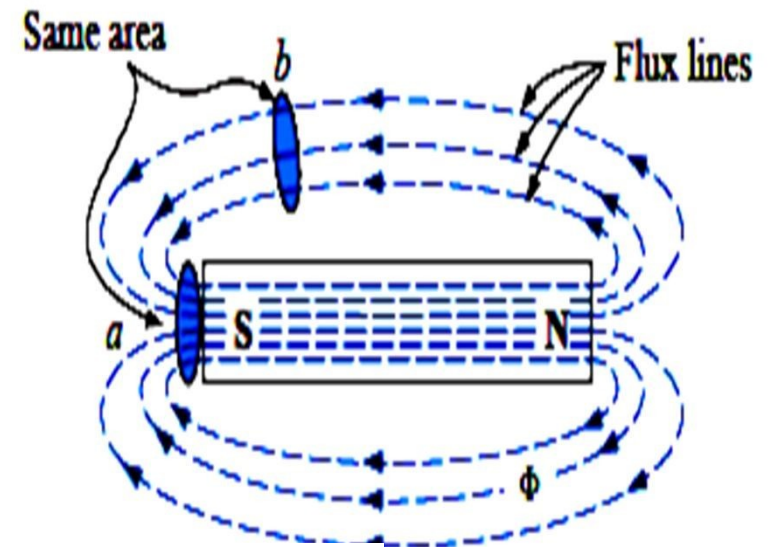
CHAPTER THREE

ELECTROMAGNETISM

MAGNETIC FIELDS

- In the region surrounding a permanent magnet there exists a magnetic field, which can be represented by **magnetic flux lines** similar to electric flux lines.
- Magnetic flux lines, however, do not have origins or terminating points as do electric flux lines but exist in **continuous loops**.
- It radiates from the north pole to the south pole, returning to the north pole through the metallic bar. as shown below.

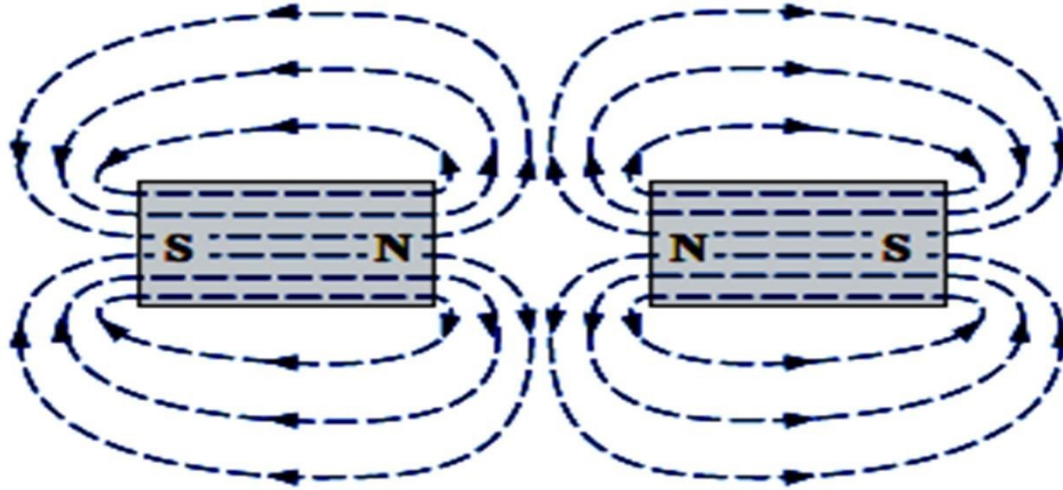
- The strength of a magnetic field in a particular region is directly related to the density of flux lines in that region.
- For example, the magnetic field strength at *a* is *twice that at b*.
- If unlike poles of two permanent



attract,
distribution will be as shown below.

magnets are
brought
together, the
magnets will
and the flux

- If like poles are brought together, the magnets will repel, and the flux distribution will be as shown.



- If a nonmagnetic material, such as glass or copper, is placed in the flux paths surrounding a permanent magnet, there will be an almost unnoticeable change in the flux distribution
- However, if a magnetic material, such as soft iron, is placed in the flux path, the flux lines will pass through the soft iron rather than

the surrounding air because the iron provides a magnetic path that is more easily established than that of air.

- ✓ Magnetic materials (materials that are attracted by magnets such as iron, nickel, cobalt, and their alloys) are called ferromagnetic materials.
- ✓ Ferromagnetic materials provide an easy path for magnetic flux.

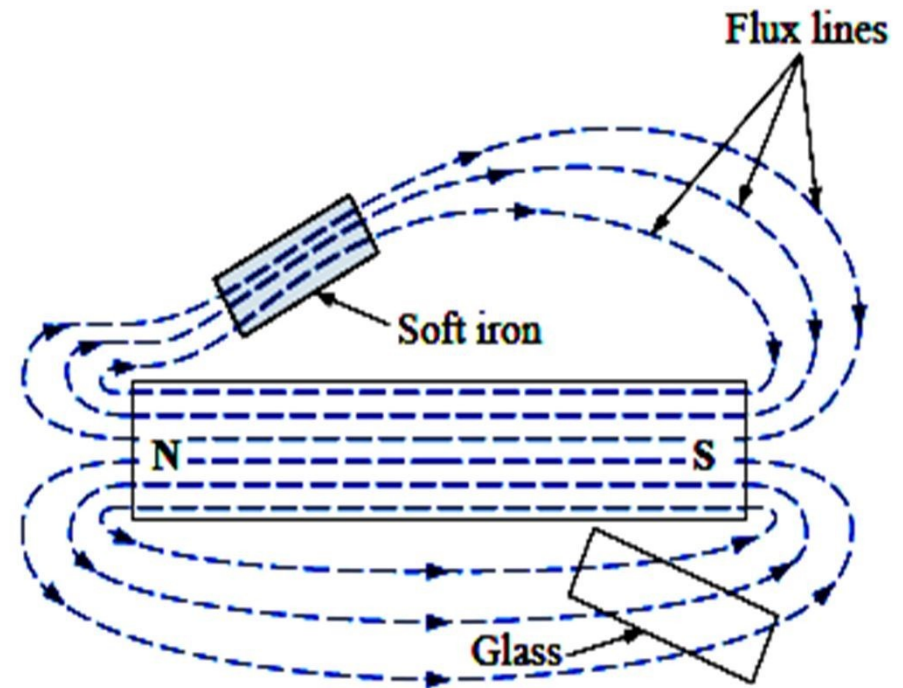


Figure. Effect of a ferromagnetic sample on the flux distribution of a permanent magnet.

MAGNETIC FLUX (ϕ)

- The group of force lines going from the north pole to the south pole of a magnet is called the magnetic flux, symbolized by ϕ (the Greek letter phi).
- The number of lines of force in a magnetic field determines the value of the flux.
- The more lines of force, the greater the flux and the stronger the magnetic field.
- The unit of magnetic flux is the weber (Wb). One weber equals 10^8 lines.
- The weber is a very large unit; thus, in most practical situations, the micro weber (μWb) is used.

MAGNETIC FLUX DENSITY (B)

- The magnetic flux density(B) is the amount of flux per unit area perpendicular to the magnetic field.
- Its symbol is B, and its SI unit is the tesla (T).
- The following formula expresses the flux density(B):
where ϕ , is the number of flux lines passing through the area A.

$$B = \frac{\Phi}{A}$$

- 1Wb = 10^8 lines
- The Gauss Although the tesla (T) is the SI unit for flux density, another unit called the gauss is occasionally used. **1T = 10^4 gauss**

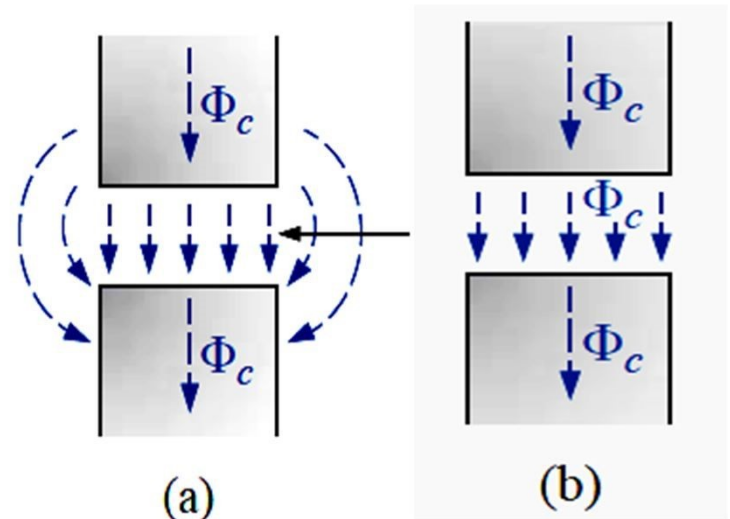
AIR GAPS

- The spreading of the flux lines outside the common area of the core for the air gap in Fig. (a) is known as fringing.
- For magnetic circuits with air gaps, fringing occurs, causing a decrease in flux density in the gap as in Figure 12–12(a).
- For our purposes, we shall neglect this effect and assume the flux distribution to be as in Fig.(b).
- The flux density of the air gap in Fig. (b) is given by:

$$B_g = \frac{\Phi_g}{A_g}$$

- For our purpose,

$$A_g = A_{\text{core}}$$



$$\Phi_g = \Phi_{\text{core}} \text{ and}$$

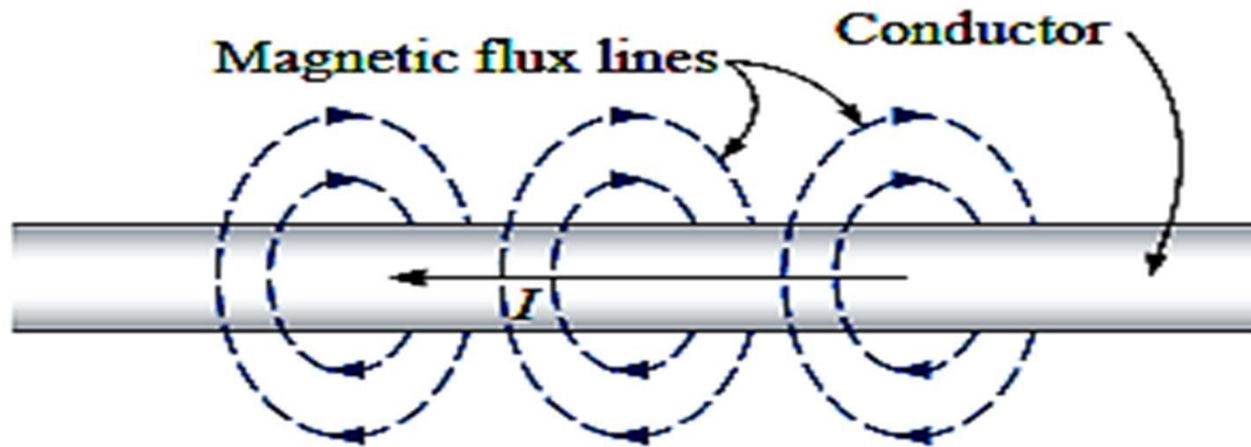
Fig. Air gaps: (a) with fringing; (b) ideal.

ELECTROMAGNETISM

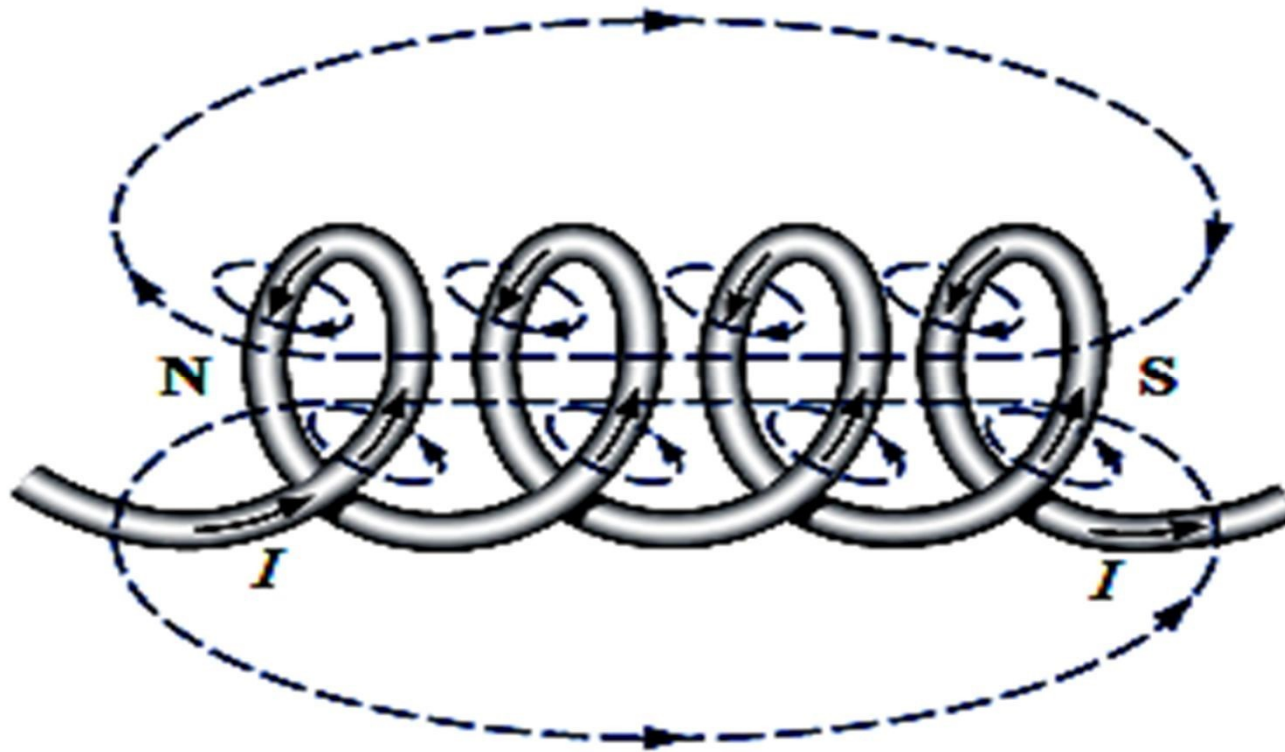
- In 1820, the Danish physicist Hans Christian Oersted discovered that the needle of a compass would deflect if brought near a current carrying conductor.
-
- **Conclusion:**
 - ✓ Since the needle is the magnetic material, there must be a magnetic field around current carrying conductor.

- ✓ A current flowing in a wire always gives rise to a magnetic field around it.
- Electromagnetism is the production of a magnetic field by current in a conductor.
- Current produces a magnetic field, called an electromagnetic field, around a conductor, as illustrated in Figure below.
- Magnetic field is present around every wire that carries an electric current.
- The invisible lines of force of the magnetic field form a concentric circular pattern around the conductor and are continuous along its length.
- The direction of the magnetic flux lines can be found by **right hand rule**; (i.e. place the thumb of the *right hand in the* direction of

conventional current flow, the fingers indicates the direction of magnetic flux lines.)



- A coil of more than one turn would produce a magnetic field that would exist in a continuous path **through** and **around** the coil.



Wednesday, August 08, 2012

- The flux distribution of the coil is quite similar to that of the permanent magnet.
- But the difference b/n the two-flux distribution is that:
 - ✓ The flux lines are more **concentrated** for the permanent magnet than for the coil.
 - ✓ Hence, the coil has a **weaker** field strength than the permanent magnet.
- The field strength of the coil can be increased by inserting certain **ferromagnetic materials**, such as iron, steel, or cobalt, within the coil to increase the flux density within the coil, hence we have devised electromagnet.

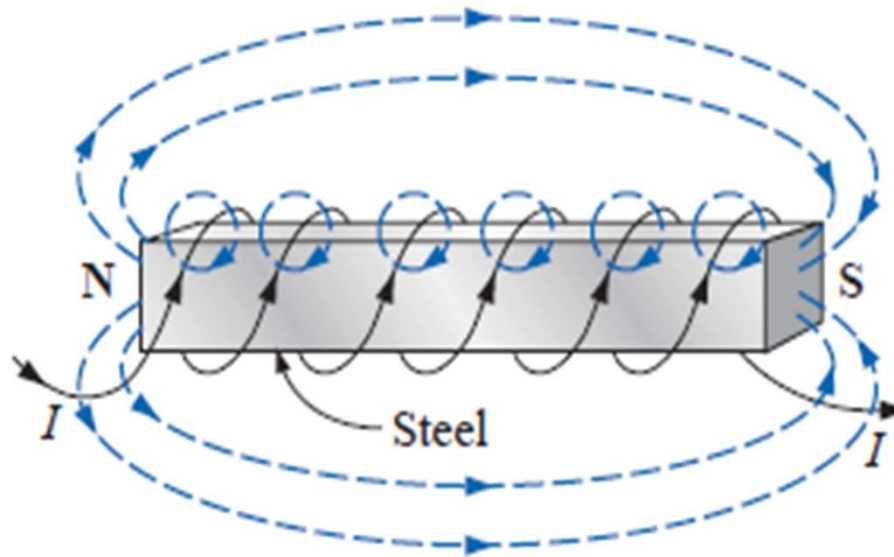
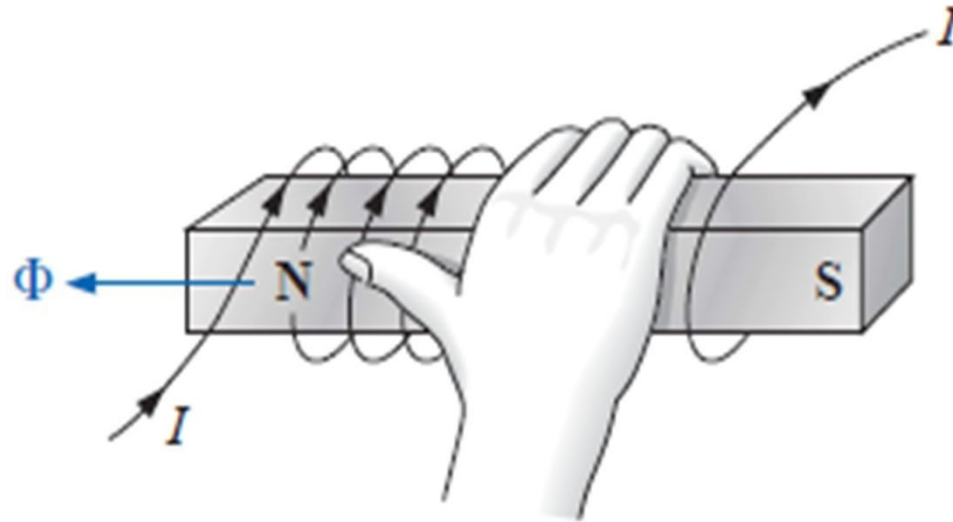


fig. Electromagnet

Note: the differences b/n electromagnet and permanent magnet are:

- ✓ By changing the component values (such as current or number of turns) electromagnet's field strength that can be varied.
- ✓ The current has to pass through the coil of electromagnet in order for magnetic flux to develop.
- ✓ There is no need for the coil or current in the permanent magnet.

- **The direction of flux lines can be determined for the electromagnet, as follows;**
 - ✓ **Place the fingers of the right hand in the direction of current flow around the core.**
 - ✓ **The thumb will then point in the direction of the north pole of the induced magnetic flux, as shown below.**



PERMEABILITY

- The strength of the magnet will vary in accordance with the core used, this variation in strength is due to the greater or lesser number of flux lines passing through the core.
- Materials in which flux lines can readily be set up are said to be magnetic and to have high permeability(μ).

- The permeability (μ) of a material, therefore, is a measure of the ease with which magnetic flux lines can be established in the material.
- It is similar in many respects to conductivity in electric circuits.
- The symbol of permeability is μ (the Greek letter mu), and its value varies depending on the type of material.
- The permeability of a vacuum (μ_0) is used as a reference.

i.e.
$$\mu_0 = 4\pi \times 10^{-7} \frac{\text{Wb}}{\text{A} \cdot \text{m}}$$

- Magnetic materials (materials that are attracted by magnets such as iron, nickel, cobalt, and their alloys) are called ferromagnetic materials.

- Ferromagnetic materials typically have permeability hundreds of times larger than that of a vacuum, indicating that a magnetic field can be set up with relative ease in these materials.
- Ferromagnetic materials include iron, steel, nickel, cobalt, and their alloys.
- The relative permeability (μ_r) of a material is the ratio of its absolute permeability to the permeability of a vacuum(μ_0).

$$\mu_r = \frac{\mu}{\mu_0}$$

- Depending on the μ_r value, materials can be classified as follows:
 - a) **Diamagnetic materials:** have permeability slightly less than μ_0 . (hence, $\mu_r < 1$)
e.g. Copper, Mercury, Zinc, gold, silver, etc
 - b) **Paramagnetic materials:** have permeability slightly greater than μ_0 (hence, $\mu_r > 1$ /slightly/) e.g. aluminum, platinum, manganese, chromium
 - c) **Ferromagnetic materials:** are materials with very high permeability,(hundreds and even thousands times that of free space.) hence; $\mu_r \gg 1$
e.g.: Iron, nickel, cobalt, steel and their alloys.

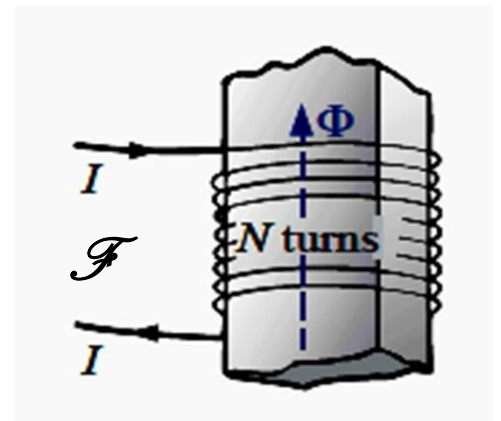
MAGNETO MOTIVE FORCE (mmf)

- As you have learned, current in a conductor produces a magnetic field, and the cause of a magnetic field is called the magnetomotive force (mmf).
- The unit of mmf, the ampere-turn (At), is established on the basis of the current in a single loop (turn) of wire.
- The formula for magnetomotive force (mmf) is given by

$$\mathcal{F} = NI$$

where $\mathcal{F}$ is the magnetomotive force, N is the number of turns of wire, and I is the current in amperes.

- mmf($\mathcal{F}$) is proportional to the product of the number of turns(N) around the core (in which the flux is to be established) and the current through the turns of wire.



- Increase in the number of turns or the current through the wire will result in an increased driving force on the system.

RELUCTANCE(R)

- The resistance of a material to the flow of charge (current) is determined for electric circuits by the equation:

$$R = \rho \frac{l}{A}$$

- The opposition to the establishment of a magnetic field in a material is called Reluctance (R).
- Reluctance in magnetic circuits is analogous to resistance in electric circuits.
- The value of reluctance is directly proportional to the length (l) of the magnetic path and inversely proportional to the permeability

(μ) and to the cross-sectional area (A) of the material, as expressed by the following equation:

$$R = \frac{l}{\mu A}$$

= is the reluctance
 l = is the length of the magnetic path
 and

A = is the cross-sectional area.

- The unit of reluctance can be derived as follows:

$$\mathcal{R} = \frac{l}{\mu A} = \frac{\cancel{\text{m}}}{(\text{Wb/At} \cdot \cancel{\text{m}})(\text{m}^2)} = \frac{\text{At}}{\text{Wb}}$$

- At/Wb is ampere-turns/weber which is the unit of reluctance. ▪ Note that:-
 - ✓ The increase in area will result in a reduction in reluctance and an increase in the desired result (flux).
 - ✓ For an increase in length, reluctance will increase, hence the no. of flux passing through an area will reduce.

- ✓ Materials with high permeability, such as the ferromagnetic, have very small reluctances and will result in an increased measure of flux through the core. **example:**

- 1) Mild steel has a relative permeability of 800. Calculate the reluctance of a mild steel core that has a length of 10 cm and has a cross section of 1.0 cm X 1.2 cm.

OHM'S LAW FOR MAGNETIC CIRCUITS

- There is a broad similarities between electric & magnetic circuits analysis. The analogous quantities are given below:

Electric Circuits	Magnetic Circuits
E	$\mathcal{F}$
I	Φ
R	$\mathcal{R}$

- The relationship between flux, mmf, and reluctance is

$$\Phi = \mathcal{F}/\mathcal{R} \quad (\text{Wb})$$

- Which is called ohm's law for magnetic ckt.
- This relationship is similar to Ohm's law and is depicted symbolically in Figure below.
(Remember however that flux, unlike electric current, does not flow.)

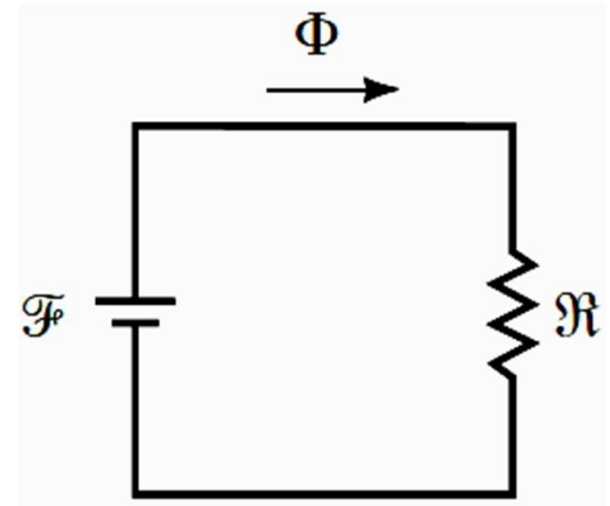


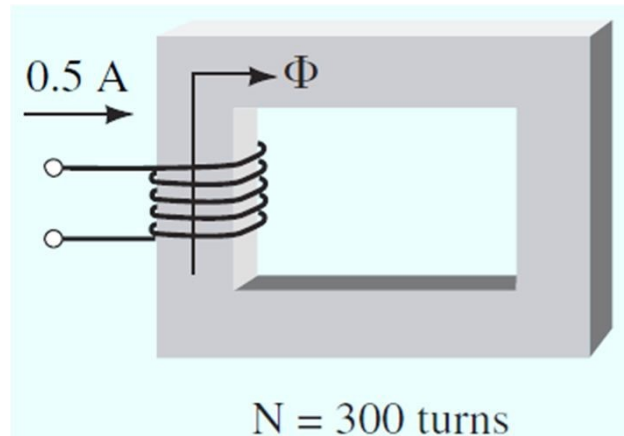
fig. Electric circuit analogy of a magnetic circuit.

- For magnetic circuit;
 - ✓ The effect desired is the flux (ϕ)
 - ✓ The cause is the magneto motive force (*mmf*) $\mathcal{F}$, which is the external force (or “pressure”) required to set up the magnetic flux lines within the magnetic material.

- ✓ The opposition to the setting up of the flux (ϕ) is the reluctance(R).

example

1) For Figure below, if the reluctance of the magnetic circuit is $R = 12 \times 10^4 \text{ At/Wb}$, what is the flux in the circuit?

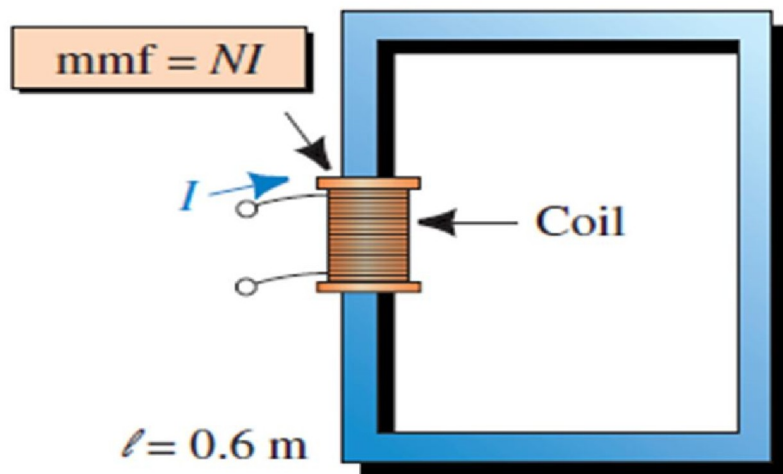


MAGNETIZING FORCE (H)

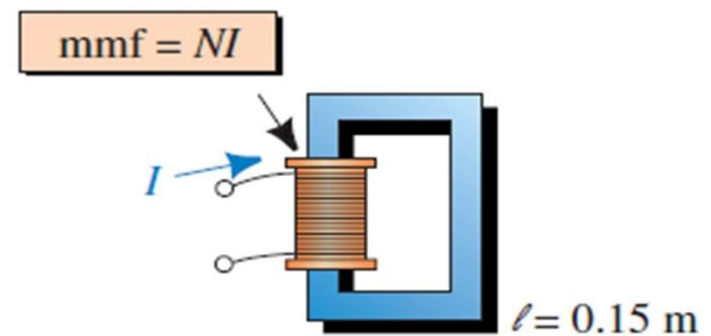
- The magneto motive force per unit length is called the magnetizing force(magnetic field intensity) (H) *and in equation form,*

$$H = \frac{\mathcal{F}}{l} = H = \frac{NI}{l}$$

- The unit of magnetizing force (H) is ampere-turns per meter (At/m)
- *To get the idea magnetic field intensity(H), suppose you apply the same mmf (600 At) to two circuits with different path lengths in Figure below.*
- *In (a), you have 600 At of mmf to “drive” flux through 0.6 m of core; in (b), you have the same mmf but it is spread across only 0.15 m of length.*



(a) A long path



(b) A short path

- ***Thus the mmf per unit length in the second case is more intense.***
- ***Based on this idea, one can define magnetic field intensity as the ratio of applied mmf to the length of path that it acts over.***

$$H = \mathcal{F}/\ell = NI/\ell \quad (\text{At/m})$$
- ***Thus,***
- ***For the circuit of Figure above (a), $H = 600 \text{ At}/0.6 \text{ m} = 1000 \text{ At/m}$, while for the circuit of (b), $H = 600 \text{ At}/0.15 = 4000 \text{ At/m}$.***
- ***Thus, in (a) you have 1000 ampere-turns of “driving force” per meter of length to establish flux in the core, whereas in (b) you have four times as much***
- ***Rearranging the above Equation yields an important result:***

$$NI = H\ell \quad (\text{At})$$

- *In an analogy with electric circuits, the NI product is an mmf source, while the $H\ell$ product is an mmf drop.*

THE RELATIONSHIP BETWEEN B AND H

- From the above Equation, you can see that magnetizing force, H , is a measure of the flux-producing ability of the coil (since it depends on NI).
- You know that B is a measure of the resulting flux (since $B = \phi/A$). Thus, B and H are related. ▪ The relationship is

where μ is the permeability of the core

$$B = \mu H$$

- For a particular H , the greater the permeability, the greater will be the induced flux density.

- The permeability of free space is $\mu_0 = 4\pi \times 10^{-7}$. For all practical purposes, the permeability of air and other nonmagnetic materials is the same as for a vacuum. ■ Thus, in air gaps,

$$B_g = \mu_0 H_g = 4\pi \times 10^{-7} \times H_g$$

B-H CURVES

- *For ferromagnetic materials(such as **iron, nickel, cobalt, and their alloys**), μ is not constant but varies with flux density and there is no easy way to compute it. In reality, however, it isn't μ that you are interested in:*
- *What you really want to know is, given B , what is H , and vice versa. A set of curves, called B-H or magnetization curves, provides this information. (These curves are obtained experimentally and are available in hand book)*

- *So that their value of B or H is obtained from the B - H curves, and A separate curve is required for each material.*

AMPERE'S CIRCUITAL LAW

- Ampere's law was determined experimentally and is a generalization of the relationship $\mathcal{F} = NI = Hl$ that we developed earlier.
- Ampere showed that the algebraic sum of mmfs around a closed loop in a magnetic circuit is zero, regardless of the number of sections or coils.
- That is,. $\sum_{\text{C}} \mathcal{F} = 0$
- This can be rewritten as

$$\sum_{\bigcirc} NI = \sum_{\bigcirc} H\ell \quad \text{At}$$

- which states that the sum of applied mmfs(NI) around a closed loop equals the sum of the mmf drops($H\ell$).
- The summation is algebraic and terms are additive or subtractive, depending on the direction of flux and how the coils are wound.
- The equation for the mmf drop across a portion of a magnetic circuit can be found by using:

$$\mathcal{F} = \Phi \mathcal{R} \quad (\text{At})$$

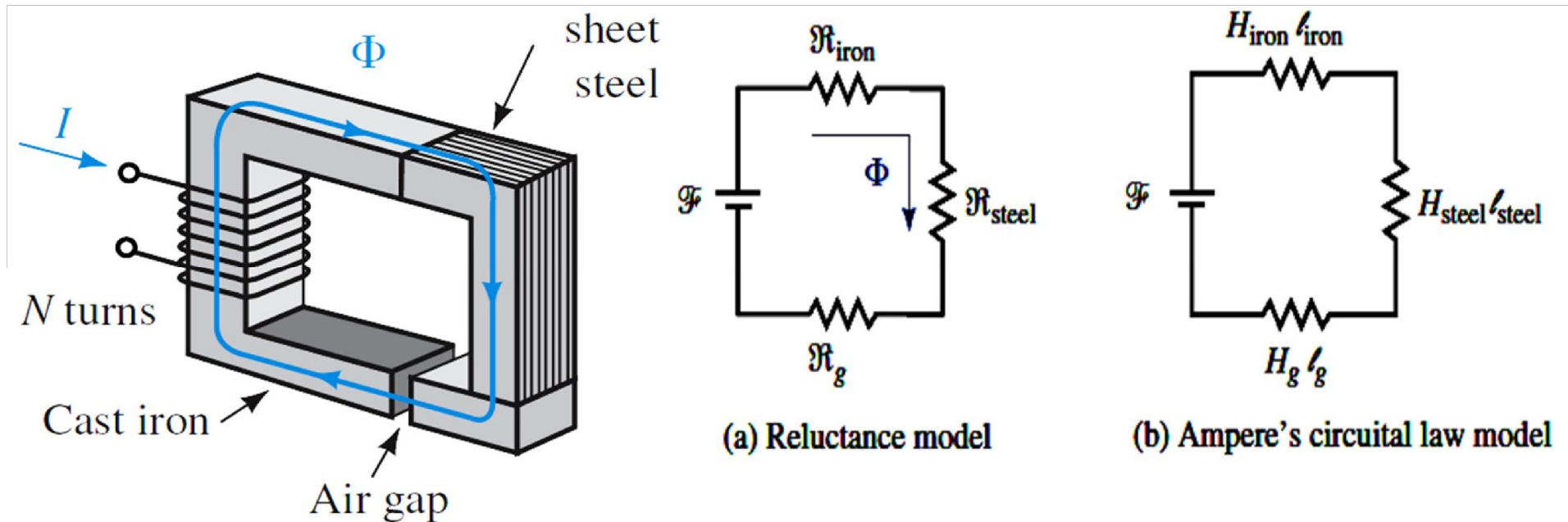
where ϕ is the flux passing through a section of the magnetic circuit and R is the reluctance of that section.

- A more practical equation for the mmf drop is:

$$\boxed{\mathcal{F} = Hl} \quad (\text{At})$$

where H is the magnetizing force on a section of a magnetic circuit and l is the length of the section.

- To illustrate, consider the following Figure .



Here,

$$NI - H_{\text{iron}} \ell_{\text{iron}} - H_{\text{steel}} \ell_{\text{steel}} - H_g \ell_g = 0$$

thus,

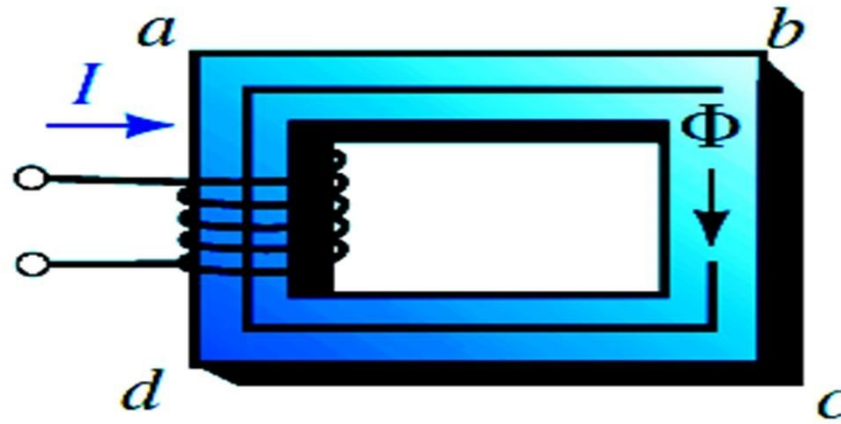
$$\underbrace{NI}_{\text{Impressed mmf}} = \underbrace{H_{\text{iron}} \ell_{\text{iron}} + H_{\text{steel}} \ell_{\text{steel}} + H_g \ell_g}_{\text{sum of mmf drops}}$$

- which states that the applied mmf NI is equal to the sum of the $H\ell$ drops around the loop.
- We have two magnetic circuit models (from the two ampere's circuital law model is best to solve the problem)

Example

- 1) A circuit consists of one coil, a section of iron, a section of steel, and two air gaps (of different sizes). Draw the Ampere's law model.
- 2) If the core of Figure below has $\mu_r = 250$ and $\phi = 0.1 \times 10^{-3}$ Wb, then;
 - a) what is the coil current?
 - b) Draw the Ampere's law model.

c) Determine the mmf source.



$$\ell_{abcd} = 0.25 \text{ m}$$

$$N = 500 \text{ turns}$$

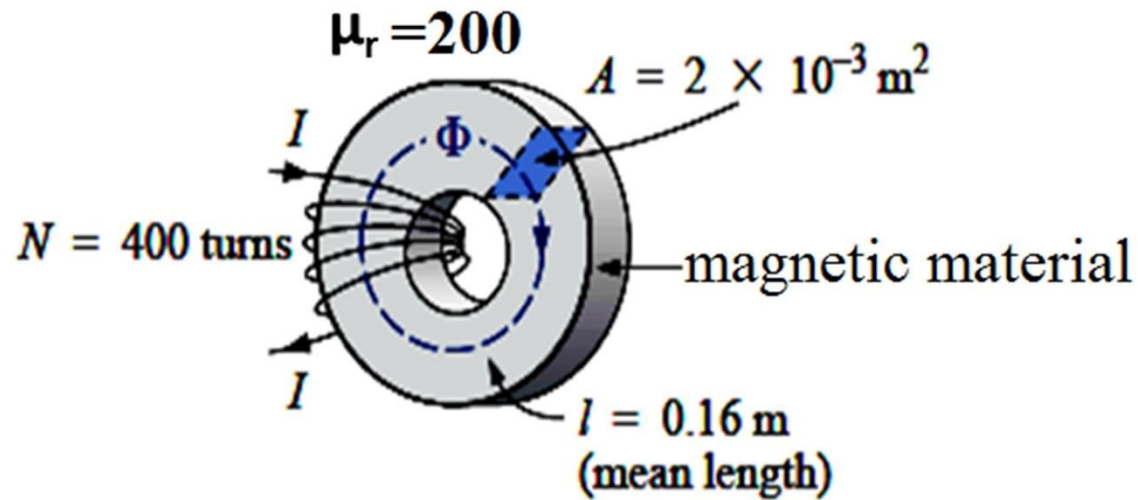
$$A = 0.2 \times 10^{-3} \text{ m}^2$$

Example

3) If the core of Figure below has $\mu_r = 200$ and $\phi = 4 \times 10^{-4} \text{ Wb}$, then;

- Find magnetizing force(H).
- The flux density(B).
- Determine the permeability of the material μ

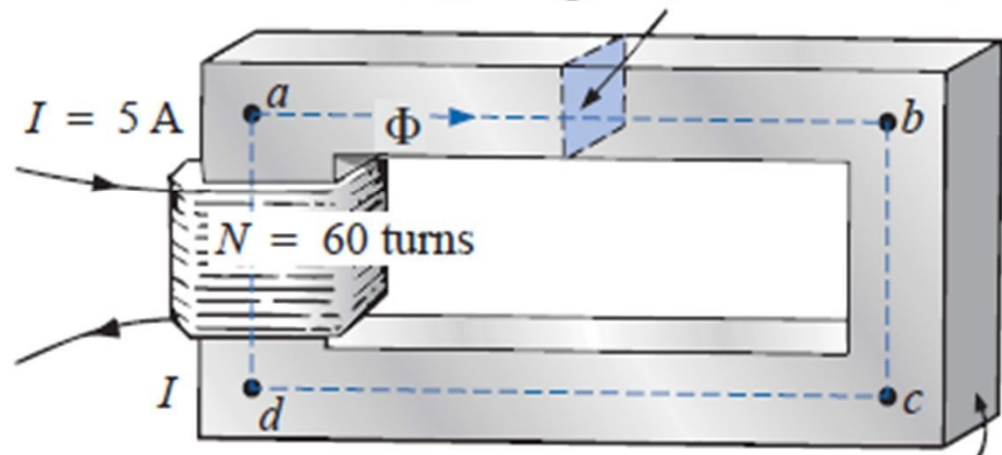
- d) Find the value of I required to develop a magnetic flux
- e) Determine the mmf source.



- 4) For the following the magnetic circuit that has $\mu_r = 150$,
- a) Calculate the magnetic flux
 - b) Find magnetizing force(H).
 - c) The flux density(B).
 - d) Determine the permeability of the material μ

$$\mu_r = 150$$

$$A \text{ (throughout)} = 2 \times 10^{-4} \text{ m}^2$$



$$l_{abcd} = 0.3 \text{ m}$$

magnetic material